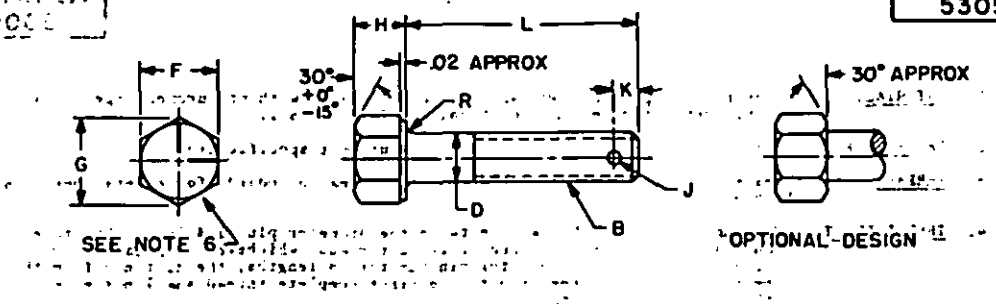
FED. SUP. CLASS
5305




| | NOMINAL SIZE | | 1/4 | 5/16 | 3/8 | 7/16 | 1/2 | 5/8 | 3/4 | 7/8 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| B | THREAD UNC-2A | | 20 | 18 | 16 | 14 | 13 | 11 | 10 | 9 | 8 |
| D | BODY DIAMETER | MAX | .2500 | .3125 | .3750 | .4375 | .5000 | .6250 | .7500 | .8750 | 1.0000 |
| | | MIN | .2450 | .3065 | .3690 | .4305 | .4930 | .6170 | .7410 | .8660 | .9900 |
| F | WIDTH ACROSS FLATS | MAX | .4375 | .5000 | .5625 | .6250 | .7500 | .9375 | 1.1250 | 1.3125 | 1.5000 |
| | | MIN | .4280 | .4890 | .5510 | .6120 | .7360 | .9220 | 1.1000 | 1.2850 | 1.4690 |
| G | WIDTH ACROSS CORNERS | MAX | .505 | .577 | .650 | .722 | .866 | 1.083 | 1.299 | 1.516 | 1.732 |
| | | MIN | .488 | .557 | .628 | .698 | .840 | 1.051 | 1.254 | 1.465 | 1.675 |
| H | HEAD HEIGHT | MAX | .163 | .211 | .243 | .291 | .323 | .403 | .483 | .563 | .627 |
| | | MIN | .150 | .195 | .226 | .272 | .302 | .378 | .455 | .531 | .591 |
| J | HOLE DIAMETER | +.006 -.002 | .076 | .076 | .107 | .107 | .107 | .140 | .140 | .140 | .140 |
| K | | ±.010 | .109 | .125 | .156 | .156 | .172 | .188 | .219 | .250 | .250 |
| R | RADIUS | MAX | .025 | .025 | .025 | .025 | .025 | .045 | .045 | .065 | .095 |
| | | MIN | .015 | .015 | .015 | .015 | .015 | .020 | .020 | .040 | .060 |
| MIN TENSILE STRENGTH - LBS (B) | | | 2,220 | 3,660 | 5,420 | 7,420 | 9,900 | 15,800 | 23,150 | 32,300 | 42,400 |

| L LENGTH | TOLERANCE SIZE 3/4 AND UNDER | TOLERANCE SIZE 7/8 AND OVER | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. | DASH NO. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| .375 | +0 -.031 | --- | 1 | 27 | | | | | | | |
| .438 | | | 2 | 28 | | | | | | | |
| .500 | | | 3 | 29 | 55 | | | | | | |
| .562 | | | 4 | 30 | 56 | | | | | | |
| .625 | | | 5 | 31 | 57 | | | | | | |
| .750 | | | 6 | 32 | 58 | 84 | 110 | 133 | | | |
| .875 | | | 7 | 33 | 59 | 85 | 111 | 134 | | | |
| 1.000 | | | 8 | 34 | 60 | 86 | 112 | 135 | 157 | | |
| 1.250 | +0 -.062 | +0 -.125 | 9 | 35 | 61 | 87 | 113 | 136 | 158 | 182 | 206 |
| 1.500 | | | 10 | 36 | 62 | 88 | 114 | 137 | 159 | 183 | 207 |
| 1.750 | | | 11 | 37 | 63 | 89 | 115 | 138 | 160 | 184 | 208 |
| 2.000 | | | 12 | 38 | 64 | 90 | 116 | 139 | 161 | 185 | 209 |
| 2.250 | +0 -.094 | +0 -.188 | 13 | 39 | 65 | 91 | 117 | 140 | 162 | 186 | 210 |
| 2.500 | | | 14 | 40 | 66 | 92 | 118 | 141 | 163 | 187 | 211 |
| 2.750 | | | 15 | 41 | 67 | 93 | 119 | 142 | 164 | 188 | 212 |
| 3.000 | | | 16 | 42 | 68 | 94 | 120 | 143 | 165 | 189 | 213 |
| 3.250 | | | 17 | 43 | 69 | 95 | 121 | 144 | 166 | 190 | 214 |
| 3.500 | | | 18 | 44 | 70 | 96 | 122 | 145 | 167 | 191 | 215 |
| 3.750 | | | 19 | 45 | 71 | 97 | 123 | 146 | 168 | 192 | 216 |
| 4.000 | | | 20 | 46 | 72 | 98 | 124 | 147 | 169 | 193 | 217 |
| 4.250 | | | 21 | 47 | 73 | 99 | 125 | 148 | 170 | 194 | 218 |
| 4.500 | | | 22 | 48 | 74 | 100 | 126 | 149 | 171 | 195 | 219 |
| 4.750 | | | 23 | 49 | 75 | 101 | 127 | 150 | 172 | 196 | 220 |
| 5.000 | | | 24 | 50 | 76 | 102 | 128 | 151 | 173 | 197 | 221 |
| 5.500 | | | | | | | 129 | 152 | 174 | 198 | 222 |
| 6.000 | | | | | | | 130 | 153 | 175 | 199 | 223 |

NOTE: For notes see sheet 2.

(B) For changes see sheets 1 and 2.

APPROVED 16 JUL 1958. REVISED (A) 14 OCT. 1968 (B) 1 MAY 1969

REVIEWER: AT, AV, IS, NSA, 05, 85
USER: AS, CL, MC, ME, MI, YD

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC
Other Coat SH
82

TITLE
SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), SHANK DRILLED FOR COTTER PIN, STEEL, CORROSION-RESISTING, PASSIVATED, UNC-2A

MILITARY STANDARD
MS51109

PROCUREMENT SPECIFICATION
FF-S-85

SUPERSEDES

SHEET 1 OF 2

DD FORM 672-1 (Continuation)

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FED. SUP CLASS
5305

NOTES:

(B) 1. MATERIAL: Corrosion-resisting steel, 400 series, in accordance with procurement specification. Minimum ultimate tensile strength: 70,000 p.s.i.

2. PROTECTIVE COATING: Passivated in accordance with procurement specification.

3. THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H28.

4. THREAD LENGTH: Minimum thread length shall be twice the diameter plus 1/4 inch. The tolerance shall be plus 3/16 inch or 2-1/2 threads, whichever is greater. On screws that are too short for minimum thread lengths, the distance from the bearing surface of the head to the first complete thread shall not exceed the length of 2-1/2 threads.

5. DIMENSIONS: All dimensions are in inches, unless otherwise specified.

(B) 6. HEAD MARKING: Heads shall be marked in accordance with MIL-HDBK-131. Raised markings are preferred, stampings are acceptable.

7. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS51109-1.

8. Referenced documents shall be of the issue in effect on date of invitations for bid.

9. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

REVIEWER:AT, AV, IS, NSA, OS, 85
USER:AS, CL, MC, ME, MI, YD

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 16 JUL 1958 REVISED (A) 14 OCT 1966 (B) FOR CHANGES SEE SHEETS 1 AND 2.

P.A.: MC
Other Cust.: SH
1-62

TITLE
SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), SHANK DRILLED FOR COTTER PIN, STEEL, CORROSION-RESISTING, PASSIVATED, UNC-2A

MILITARY STANDARD
MS51109

PROCUREMENT SPECIFICATION
FF-S-85

SUPERSEDES:

SHEET 2 OF

DD FORM 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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